

Coffee Shop Sales Analysis

SQL Project | Harshit

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MySQL | Window Functions | KPI Analysis | Time-Series | Data Cleaning

01 — PROJECT OVERVIEW

What Is This Project?

This project performs a complete end-to-end sales analysis for a multi-location coffee shop chain using MySQL. Starting from a raw transactional dataset, the work covers data cleaning, type correction, and a comprehensive suite of KPI queries designed to support real business decision-making.

02 — DATASET

About the Data

Total Records	149,000+ transactions
Locations	3 store branches
Time Period	Multiple months (2023)
Key Columns	transaction_id, transaction_date, transaction_time, transaction_qty, unit_price, product_category, product_type, store_location
Database	MySQL (coffee_shop_db)
Table Name	coffee_shop

03 — BUSINESS QUESTIONS

What We Set Out to Answer

- What are the total sales, orders, and quantity sold for each month?
- How does this month's performance compare to last month (MoM growth)?
- Which store location drives the most revenue?
- When are customers buying — what hour and day of week peaks the most?
- Which product categories and individual items generate the highest revenue?
- On which days is performance above or below the monthly average?

04 — DATA CLEANING

Cleaning Steps Performed

- transaction_date from VARCHAR to DATE using STR_TO_DATE()
- transaction_time from VARCHAR to TIME
- transaction_id column (contained BOM/encoding characters)
- Handled NULL values with conditional WHERE clauses during UPDATE operations
- Disabled safe update mode (SET sql_safe_updates = 0) for bulk cleaning

05 — METHODOLOGY

Analysis Pipeline

Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
Data Import	Data Cleaning	KPI Queries	MoM Analysis	Segmentation	Benchmarking
Raw CSV imported into MySQL	Fix types, NULLs, rename cols	Sales, orders & quantity/month	LAG() for growth %	Location, product, time	Daily vs. monthly avg

06 — ANALYSIS MODULES

What Was Built

#	Analysis Module	Description	SQL Techniques
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01	Monthly KPI Summary	Total Sales, Orders & Quantity — formatted output for any selected month	SUM(), COUNT(), CONCAT()
02	Month-on-Month Growth	% change in sales, orders, and quantity vs. the previous month	LAG(), OVER(), Window Function
03	Weekday vs. Weekend Sales	Revenue segmentation to identify if weekdays or weekends drive more business	DAYOFWEEK(), CASE WHEN
04	Sales by Store Location	Ranked revenue comparison across all 3 branches for any given month	GROUP BY, ORDER BY
05	Daily Sales vs. Avg Benchmark	Tags each day as Above / Below / Average using a dynamic window AVG	AVG() OVER(), Subquery, CASE
06	Sales by Hour & Day of Week	Identifies peak hour (8 AM) and best-performing days using time functions	HOUR(), DAYOFWEEK(), CASE WHEN
07	Product Category & Top 10	Ranks all 9+ product categories; drills into top items within each	GROUP BY, LIMIT, WHERE filter

07 — KEY FINDINGS

What the Data Revealed

8 AM Peak Sales Hour The highest transaction volume and revenue consistently occurs at 8 AM, driven by the morning commute rush.	COFFEE Top Product Category Coffee is the #1 revenue-generating product category, ranking first in total sales across all months analyzed.	WEEKDAYS Weekday Traffic Dominates Weekday sales significantly outperform weekends, indicating the customer base is largely made up of daily commuters.	LOCATION GAP Store Performance Gap One store location consistently lags behind the other two — a clear opportunity for targeted operational improvement.
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08 — TECHNICAL SKILLS

SQL Techniques Demonstrated

✓ LAG() + OVER() — Window Function	✓ Subqueries — Nested Queries
✓ STR_TO_DATE() — Data Cleaning	✓ ALTER TABLE — Schema Management
✓ AVG() OVER() — Window Aggregation	✓ GROUP BY + ORDER BY — Aggregation & Sorting
✓ CASE WHEN — Conditional Logic	✓ DAYOFWEEK() — Time Segmentation
✓ HOUR() / MONTH() — Date & Time Functions	✓ CONCAT() + ROUND() — Output Formatting

09 — HOW TO RUN

Running This Project

- Import your CSV dataset into MySQL as a table named coffee_shop
- Run the full script: Coffee_Sales_Analysis.sql
- Adjust the MONTH() filter value to explore different months (1 = Jan, 3 = Mar, etc.)
- Each analysis section is modular — sections can be run independently

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